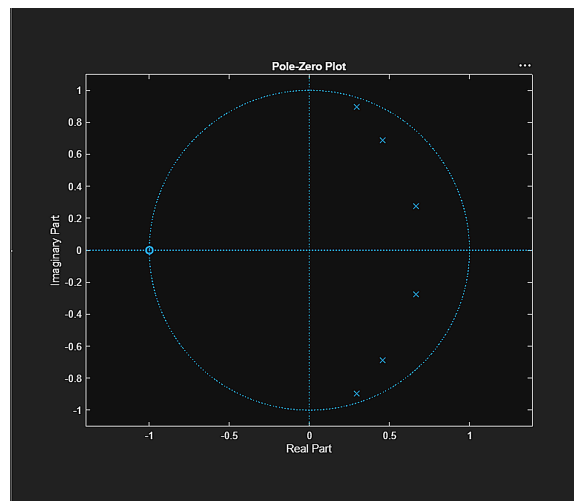
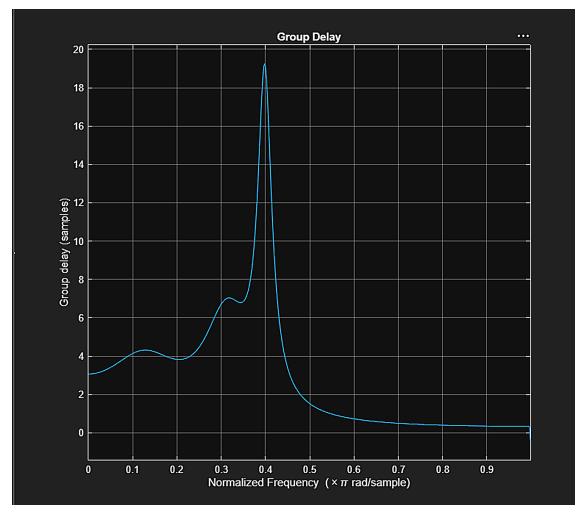
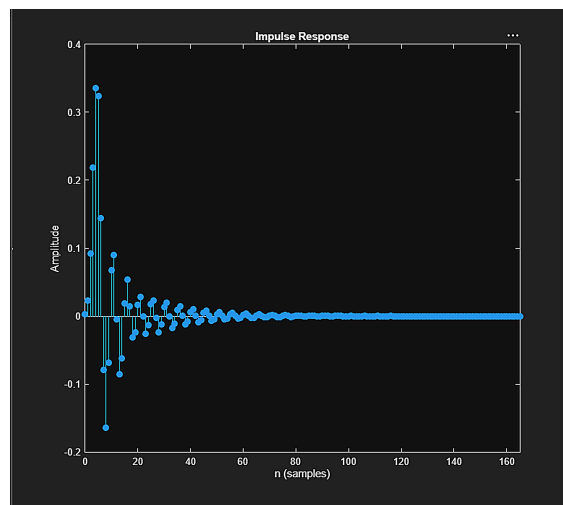
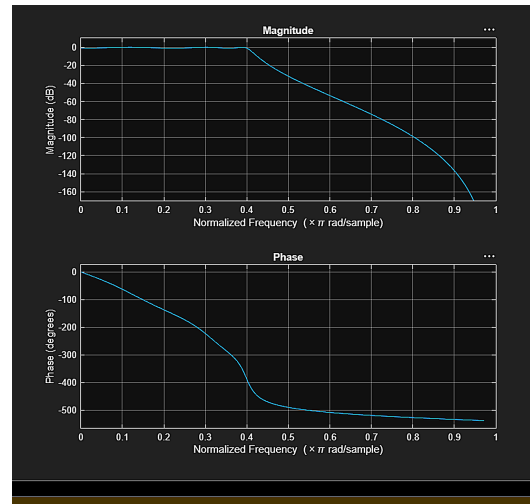


EX NO:2 Chebyshev IIR Filter

2.1 Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

```
1 clc; clear;
2
3 Np=8; Mp=55; Rp=1; Ap=40;
4
5 [b,m] = cheb1ord(Np,Mp,Rp,Ap);
6 [b,p] = cheb1z(b,m);
7
8 fprintf('Filter order = %d\n',Np);
9 fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Mp);
10
11 if all(abs(real(p)) < 1)
12     disp('System is STABLE');
13 else
14     disp('System is UNSTABLE');
15 end
16
17 % ----- DISPLAY -----
18 figure; freqz(b,p); % Magnitude + Phase
19 figure; impz(b,p); % Impulse Response
20 figure; grpdelay(b,p); % Group delay
21 figure; pzplane(b,p); % Pole-zero
22
23 Command Window
24
25 New to MATLAB? See resources for Getting Started
26
27 Filter order = 6
28 Cutoff Frequency = 0.40 * pi rad/sample
29 System is STABLE
30
31 >>
```



2.2 Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

```
1  clc;
2  clear;
3  close all;
4
5  % Specifications
6  Rp = 1; % Passband Ripple (dB)
7  Rs = 40; % Stopband Attenuation (dB)
8  Wp = 0.5; % Passband Edge
9
10
11
12  Ws = 0.35; % Stopband Edge
13
14  % Find Filter Order
15  [n,Mh] = cheb1ord(Wp,Ws,Rp,Rs);
16
17  % Design High-Pass Filter
18  [b,a] = cheby1(n,Rp,Mh,'high');
19
20  % Frequency Response
21  figure;
22  freqz(b,a);
23  title('Frequency Response');
24
25  % Impulse Response
26  figure;
27  impz(b,a);
28  title('Impulse Response');
29
30  % Group Delay
31  figure;
32  grpdelay(b,a);
33  title('Group Delay');
34
35  % Pole-Zero Plot
36  figure;
37  zplane(b,a);
38  title('Pole-Zero Plot');
39
40  fprintf('Filter Order = %d\n',n);
41  fprintf('Cutoff Frequency = %f\n',Wp);
42
43
44 Command Window
45
46 New to MATLAB? See resources for Getting Started
47
48 Filter Order = 6
49 Cutoff Frequency = 0.50000
50 >>
```

